

Scale-Up of Soluble PhaZ Hydrolase Production from 2-L to 30-L Fermentation Using Fixed Aeration and Agitation

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ABSTRACT

Production of the fictional hydrolase PhaZ was transferred from a 2-L stirred vessel to a 30-L fermenter while maintaining fixed agitation and aeration settings rather than dissolved-oxygen control. Biomass increased from 18.4 to 21.7 g wet cell weight per liter, whereas soluble PhaZ yield decreased from 8.2 to 6.3 mg/L. The insoluble fraction increased and the final acetate concentration rose from 1.1 to 3.8 g/L. Specific activity of purified soluble PhaZ was unchanged. The scale-up therefore increased biomass but reduced soluble product formation, consistent with a mass-transfer limitation that was not directly measured because the production vessel lacked a dissolved-oxygen probe.

Keywords: fermentation, scale-up, soluble expression, fixed agitation, PhaZ hydrolase

INTRODUCTION

Scale-up can preserve total biomass while changing the balance between soluble and insoluble recombinant protein. Fixed agitation and airflow are common in legacy production vessels that lack reliable dissolved-oxygen measurement.

This study compared a controlled 2-L process with a 30-L fixed-setting process for PhaZ. The primary outcomes were wet cell weight, soluble yield, insoluble fraction, acetate, and specific activity. The work did not calculate a volumetric oxygen-transfer coefficient and did not claim direct evidence of oxygen limitation.

MATERIALS AND METHODS

Strain and medium

A synthetic *E. coli* production strain carrying His6-PhaZ was grown in buffered rich medium. Induction occurred at an optical density of 7.0 using 0.05 mM IPTG, followed by 16 h at 18 C.

Fermentation conditions

The 2-L vessel used 1.4 L working volume, 700 rpm, and 1.0 vvm. Dissolved oxygen was monitored but not controlled by a cascade. The 30-L vessel used 24 L working volume, 300 rpm, and 0.5 vvm. The larger vessel had no functional dissolved-oxygen probe; agitation and aeration remained fixed.

Recovery and assays

Wet cell weight was measured after centrifugation. Soluble and insoluble PhaZ were quantified by calibrated gel densitometry. His6-PhaZ was purified by nickel affinity chromatography for specific-activity measurements using a chromogenic ester substrate. Acetate was measured enzymatically.

Parameter	2-L process	30-L process
Working volume	1.4 L	24 L
Agitation	700 rpm	300 rpm
Aeration	1.0 vvm	0.5 vvm
DO measurement	Available	Unavailable

Table 1. Fermentation settings.

RESULTS

Biomass and product distribution

Wet cell weight increased from 18.4 +/- 1.2 g/L in the 2-L vessel to 21.7 +/- 1.5 g/L in the 30-L vessel. In contrast, soluble PhaZ decreased from 8.2 +/- 0.6 to 6.3 +/- 0.7 mg/L. The fraction of PhaZ found in the insoluble pellet increased from 34% to 57%.

Metabolite accumulation

Final acetate increased from 1.1 +/- 0.2 g/L at 2 L to 3.8 +/- 0.5 g/L at 30 L. Culture pH remained within the operating range because base addition was available in both vessels.

Specific activity

Purified soluble PhaZ showed 112 +/- 9 U/mg from the 2-L process and 109 +/- 11 U/mg from the 30-L process. Thus, the soluble enzyme that was recovered had comparable catalytic quality despite the lower amount produced.

Outcome	2 L	30 L
Wet cell weight (g/L)	18.4 +/- 1.2	21.7 +/- 1.5
Soluble PhaZ (mg/L)	8.2 +/- 0.6	6.3 +/- 0.7
Insoluble PhaZ (%)	34	57
Acetate (g/L)	1.1 +/- 0.2	3.8 +/- 0.5
Specific activity (U/mg)	112 +/- 9	109 +/- 11

Table 2. Scale-up outcomes, mean +/- SD, n=3.

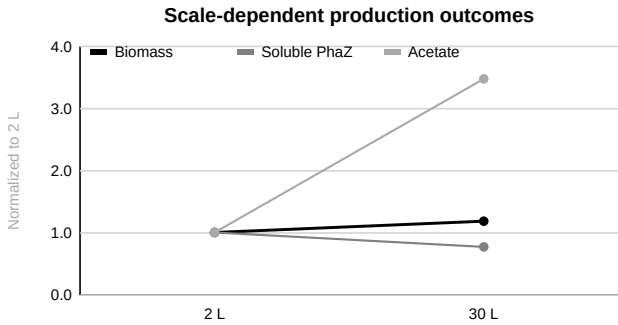


Figure 1. Biomass rose at 30 L, but soluble product fell and acetate accumulated.

DISCUSSION

The 30-L process generated more biomass but less soluble PhaZ. Increased acetate and insoluble product are compatible with reduced oxygen availability or other scale-dependent gradients, but dissolved oxygen was not measured in the larger vessel.

Unchanged specific activity indicates that the soluble fraction retained catalytic quality. The principal scale-up loss therefore occurred before purification, through reduced soluble expression rather than damage to purified enzyme.

LIMITATIONS

The absence of dissolved-oxygen data prevents direct attribution to oxygen limitation. Only one agitation and airflow setting was examined at each scale, and the experiment did not estimate kLa.

CONCLUSION

Fixed-setting scale-up from 2 L to 30 L increased biomass by 18% but reduced soluble PhaZ yield by 23%, increased the insoluble fraction, and raised acetate. The mechanism remains unresolved because dissolved oxygen was not measured at 30 L.

REFERENCES

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